

T5 Run Report: Observability & Governance

Every observed value of the recorded run, with where it comes from. Generated from the run directory; nothing typed.

T5

T5 · Production AI Engineering

5 min read · Published 2026-09-30 · Run report

Run

run	observability_governance_poc/runs/2026-09-30-recorded · preregistration sha256:08cc152fc873...
scenarios · experiments	15 scenarios covering 12 experiments, each with a concurrent background execution
model	qwen3:8b via Ollama 0.30.11; 30 calls in the scenarios, 20 in the drift probe; median 10.9 s
environment	Python 3.12.13 · OpenTelemetry SDK 1.45.0
processes	62 started (32 agent runtimes), 2 killed with SIGKILL
records	605 application log lines · 573 spans · 399 evidence events · 32 requests to the deployment API
checks	7/7 (checks.json)
expectations · predictions	68/68 · 7/7 (missed: none)
replay (2026-09-30-recorded-replay , model unreachable)	60 answers from tape, 0 misses, identical: yes; drift identical: yes
tests	31/31

Outcomes per scenario

Scenario	Recorded outcome	Production changes	Attempts reaching the API	Agent processes	Evidence events (both executions)	L0	L1	L2
e01-success	MITIGATED	1	1	1	27	10	10	13
e02-tool-failure	FAILED_NO_EFFECT	0	3	1	31	10	10	13
e03-policy-denial	DENIED	0	0	1	20	10	10	13
e04-human-rejection	REJECTED	0	0	1	22	10	10	13
e05a-crash-awaiting-approval	MITIGATED	1	1	2	28	10	10	13
e05b-crash-after-dispatch	MITIGATED	1	1	2	28	10	10	13
e06-duplicate-delivery	MITIGATED	1	2	1	29	10	10	13
e07-unexpected-capability	DENIED	0	0	1	21	10	10	13
e08a-policy-v41	MITIGATED	1	1	1	26	10	10	13
e08b-policy-v42	MITIGATED	1	1	1	27	10	10	13
e09-config-v9	MITIGATED	1	1	1	27	10	10	13
e10-restricted-data	MITIGATED	1	1	1	28	10	10	13
e11-false-success	EFFECT_NOT_OBSERVED	0	1	1	27	9	9	13
e12a-lost-response	MITIGATED	1	2	1	29	10	10	13
e12b-lost-response-no-key	MITIGATED	2	2	1	29	10	10	13

Sources: `scenarios/*/result.json`, `truth.json` (the deployment API's revisions and requests), `reconstruction.json`. L0–L2: correct answers of 13.

Reconstruction by question

Scenarios answered correctly, of 15.

	Question	L0 · logs	L1 · + traces	L2 · lineage
Q1	What triggered the execution?	15	15	15
Q2	Which principal initiated it, and for whom?	0	0	15
Q3	Which agent and version acted?	15	15	15
Q4	Which model and configuration produced the proposal?	0	0	15
Q5	Which policy and version evaluated it?	15	15	15
Q6	Was approval required?	15	15	15
Q7	Who approved or rejected it?	15	15	15
Q8	Which capability and arguments reached production?	15	15	15
Q9	How many attempts reached the production API?	15	15	15
Q10	Did the side effect occur?	15	15	15
Q11	How many times did production change?	15	15	15
Q12	Was the incident actually mitigated?	14	14	15
Q13	Can the evidence's integrity be verified?	0	0	15
Totals		L0	L1	L2
correct		149/195	149/195	195/195
wrong · partial · not recorded · ambiguous		1 · 30 · 15 · 0	1 · 30 · 15 · 0	0 · 0 · 0 · 0
key joins · timestamp joins (all scenarios)		60 · 15	75 · 0	25 · 0
most sources in one scenario		6	6	3

Source: `reports/comparison.json` . Every answer, its truth and its verdict is in the Lab Console.

Side effects in the ambiguity scenarios

	e06 duplicate delivery	e11 false success	e12a lost response	e12b no key
requests reaching the API	2	1	2	2
timeouts · replays · responses dropped	0 · 1 · 0	0 · 0 · 0	1 · 1 · 1	1 · 0 · 1
revision before → after	184 → 185	184 → 184	184 → 185	184 → 186
version after	v4.17.2	v4.18.0	v4.17.2	v4.17.2
verified by read-back	yes	no	yes	yes
transactions for the key	1	1	1	0
production changes (truth)	1	0	1	2

e12a: action `act-7b7550b9` , transaction `dtx-886a012833` , timeout after 1,503 ms, execution `exec-106b857f` , 16 evidence events for the execution, 12.6 s from start to completion; the retry followed the timeout by 208 ms. e05b (crash after dispatch): 1 attempt reconciled by lookup, 1 attempt reaching the API, 1 change. e11: the tool claimed `ROLLED_BACK` .

Governance lineage

e03 policy decision · severity	DENY · SEV-3
e07 capability · gateway refusals	<code>deployment.scale</code> · 1
e08a policy	v41 <code>sha256:6ce10d562f22...</code> · quorum 1 · verify obligation no · 1 approval
e08b policy	v42 <code>sha256:24395a0d4331...</code> · quorum 2 · verify obligation yes · 2 approvals
e01 model and configuration	qwen3:8b <code>sha256:500a1f067a9f...</code> · <code>incident-remediation@17</code> · <code>incident-agent-prod@8</code> · temperature 0 · 576+159 tokens · proposal <code>deployment.rollback payment-service v4.17.2</code>
e09 model and configuration	qwen3:8b <code>sha256:500a1f067a9f...</code> · <code>incident-remediation@18</code> · <code>incident-agent-prod@9</code> · temperature 0.3 · 575+145 tokens · proposal <code>deployment.rollback payment-service v4.17.2</code>
e10 restricted data	<code>payments.customer_transactions</code> (RESTRICTED) denied

Integrity, privacy and telemetry

evidence chains and anchors intact	15/15 · 399 events
tamper T1 edit in place	chain ok no · anchor ok yes · detected yes
tamper T2 recompute chain	chain ok yes · anchor ok no · detected yes
tamper T3 truncate	chain ok yes · anchor ok no · detected yes
tamper T4 recompute chain and anchor	chain ok yes · anchor ok yes · detected no
files scanned · canary hits · credential hits	225 · 0 · 0
spans · orphaned by SIGKILL (e05a · e05b)	573 · 13 (5 · 8)
agent metric dumps lost to SIGKILL	2
head sampling of the 15 target traces: kept at 25% · 10% · 1%	4 · 1 · 0
metric series	19
retention (config, illustrative): debug logs · ops telemetry · audit evidence	7 · 30 · 400 days

Sources: `reports/tamper.json` , `reports/telemetry.json` , `evidence/evidence-verification.json` , `config/retention.toml` .

Metrics (processes that exited cleanly, both executions)

Series	Value
<code>lineage.tool.attempts</code> result COMMITTED · REPLAYED · TIMEOUT · UNAVAILABLE	24 · 2 · 2 · 3
<code>lineage.tool.unknown_outcomes</code>	2
<code>lineage.effects.unverified</code>	2

Source: `telemetry/metrics.json` . Counts include the concurrent background executions; the two SIGKILLED processes' metrics are missing, which is the point of the telemetry row above.

Drift probe (D1)

	rollback	restart	diagnostics	no action
config v8	7	2	0	1
config v9	0	4	6	0

8 of 10 incidents changed action; 3 rollback proposals denied by policy; D01: v8 deployment.rollback , v9 pods.restart . Source: reports/drift.json .

Predictions

Prediction	Result	Evidence
P1 L2 answers all 13 questions correctly in every scenario.	held	L2 below 13/13 in: none
P2 L0 cannot verify evidence integrity (Q13) in any scenario.	held	L0 verified integrity in: none
P3 L0 reports the wrong final outcome (Q12) in e11-false-success.	held	L0 Q12 verdict in e11-false-success: WRONG
P4 L1 (logs + traces) needs fewer heuristic joins than L0 in every scenario.	held	L1 not fewer heuristic joins in: none
P5 L0 cannot say which prompt template or agent configuration produced the proposal (Q4) in any scenario.	held	L0 answered Q4 in: none
P6 The e12b control mutates production twice; e12a mutates it once.	held	mutations e12a=1, e12b=2
P7 The restricted-data canary appears in no model request, span, log line or evidence event, in any scenario.	held	canary found in: none

Source: reports/predictions.json , checked mechanically against experiments/preregistration.toml .

Run report

Evidence

Observability

Governance

PRODUCTION AI ENGINEERING · THE SERIES SO FAR

FOUNDATION

F1 · MCP Tool Sprawl

Your AI agent has 500 MCP tools. Now what?

FOUNDATION

F2 · Layered Architecture

Your agent works in a demo. Why does it break in production?

FOUNDATION

F3 · Headless AI

Your AI shouldn't live inside the UI

TRUST

T1 · Agent Identity

Who is acting, and on whose authority?

TRUST

T2 · Authorization & Policy

What is this agent actually allowed to do?

TRUST

T3 · Human-in-the-Loop

Authorized. Should it still act?

PREVIOUS

AI Control Plane

Who coordinates identity, policy, approval and budgets across every agent? (in progress)

CURRENT

T5 · Observability & Governance

Your agent changed production. Can you prove why?

NEXT

Production Agent Platform

The final reference architecture (planned)

[The full learning_map \(Start Here\)](#).

T5 · Production AI Engineering

Production AI Engineering · T5 · Run report. Every measured number here is substituted at build time from observability_governance_poc/runs/2026-09-30-recorded/facts.json; the enterprise systems are simulated and the run is deterministic. Figures carry their provenance.

- [Medium edition](#)
- [Technical deep dive](#)
- [Evidence Check](#)
- [POC README](#)
- [Run summary](#)
- [Lab Console](#)
- [Run report](#)
- [QA report](#)